

PLAYER PERFORMANCE ANALYSIS

Finding the Key Factors Behind Football Player Performance with PCA

Principal Component Analysis, Implemented From Scratch

Task 2.2 • AI Team Training '27 • Phase I

Data: FIFA 19 Complete Player Dataset (Kaggle)

Executive Summary

This report implements PCA **from scratch** (no `sklearn.decomposition.PCA`) on 18,147 FIFA 19 players, using 8 skill attributes (Finishing, ShortPassing, Dribbling, SprintSpeed, Strength, Stamina, Interceptions, StandingTackle) plus market Value. The goal: compress 9 correlated attributes into a few independent “performance archetypes.”

The first two components explain **70.2%** of all variance. **PC1 (48.0%)** is a general “overall ability” axis, loading positively on almost every attribute. **PC2 (22.2%)** cleanly separates attacking traits (Finishing, Dribbling, SprintSpeed) from defensive ones (Interceptions, StandingTackle, Strength) — an “attacker vs. defender” axis discovered with no position labels involved. The scatter plot confirms it: attackers and defenders separate along PC2 purely from statistics.

1 Data & Methodology

1.1 Preprocessing

- Loaded from the FIFA 19 dataset (18,207 players). The file is cp1252-encoded (accented names like “Modrić”, “Müller”), so read with `encoding='cp1252'`.
- *Value* was text (e.g. “€110.5M”) and parsed into a numeric euro amount.
- 60 players with no Position/skill data were dropped, leaving 18,147. Position was kept aside purely to color the final scatter plot — never used in the PCA math itself.

1.2 PCA Steps (Manual NumPy Implementation)

- Standardization** — $z = \frac{x-\mu}{\sigma}$ per feature, so Value (tens of millions) can’t dominate 0–100-scale skills.
- Covariance matrix** — $\frac{Z^T Z}{n-1}$ on standardized data (equivalent to the correlation matrix here).
- Eigen-decomposition** — `numpy.linalg.eig` on the 9×9 covariance matrix.
- Component selection** — eigenvalues sorted descending, eigenvectors reordered to match.
- Projection** — standardized data multiplied by the top 2 eigenvectors $\rightarrow$ each player’s (PC1, PC2).

2 Eigenvalues, Eigenvectors & PCA

Note: rewrite this section in your own words with your own sources before submitting — it's meant to demonstrate your own understanding.

2.1 The Core Idea

For a square matrix A , $\mathbf{v}$ is an eigenvector if A only stretches/shrinks it, never rotates it:

$$A \cdot \mathbf{v} = \lambda \cdot \mathbf{v} \quad (1)$$

where λ is the eigenvalue. For a covariance matrix, each eigenvector is a direction of independent variation in the data, and its eigenvalue is how much variance lies along that direction — large λ means high spread, small λ means almost none.

PCA finds the directions of maximum variance, each uncorrelated with the last. Those directions are exactly the covariance matrix's eigenvectors, ranked by their eigenvalues. Keeping only the top few gives a small set of uncorrelated features that preserve most of the original information — that's the dimensionality reduction.

2.2 Resources to Read & Cite

- StatQuest (Josh Starmer) — “PCA, Step-by-Step” (YouTube)
- 3Blue1Brown — “Eigenvectors and eigenvalues”, Essence of Linear Algebra (YouTube)
- Shlens, J. — “A Tutorial on Principal Component Analysis” ([arXiv:1404.1100](#))
- Strang, G. — *Linear Algebra and Learning from Data*, Wellesley-Cambridge Press

3 Results

3.1 Explained Variance

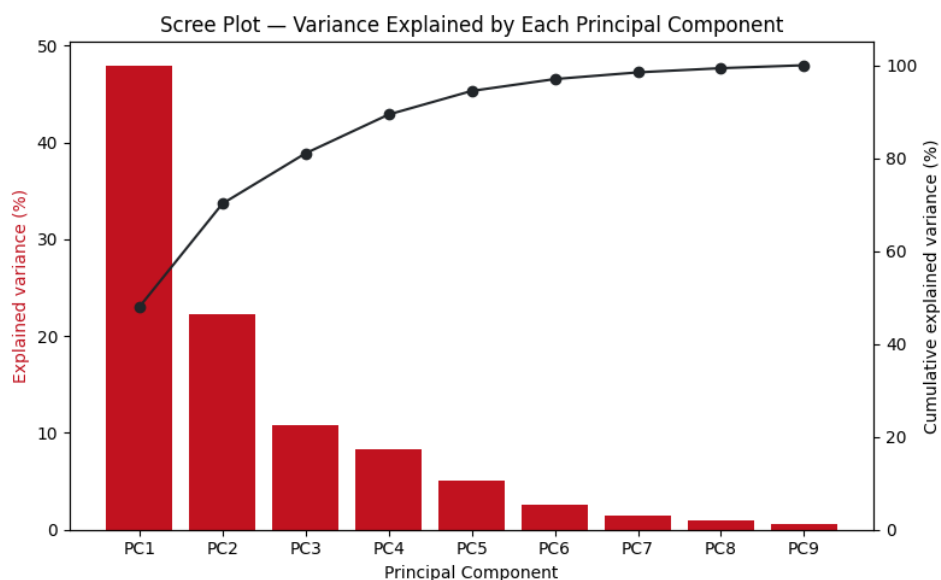


Figure 1: Variance explained per component, with cumulative variance overlaid.

The first two components already capture 70.2% of all variance — a solid basis for the 2D visualization below (5 components would be needed for 94.5%).

Table 1: Eigenvalues and Variance Explained

Component	Eigenvalue	Variance	Cumulative
PC1	4.317	48.0%	48.0%
PC2	2.002	22.2%	70.2%
PC3	0.974	10.8%	81.0%
PC4	0.752	8.4%	89.4%
PC5	0.457	5.1%	94.5%

3.2 Feature Loadings

Table 2: Principal Component Loadings

Feature	PC1	PC2
Finishing	+0.328	+0.422
ShortPassing	+0.442	+0.034
Dribbling	+0.424	+0.266
SprintSpeed	+0.342	+0.289
Strength	+0.107	-0.390
Stamina	+0.421	-0.068
Interceptions	+0.302	-0.506
StandingTackle	+0.300	-0.502
Value	+0.175	+0.052

PC1 loads positively on every attribute — an “overall quality” axis, independent of position. **PC2** splits Finishing, SprintSpeed and Dribbling (positive) from Interceptions, StandingTackle and Strength (negative): a clean, automatically-discovered attacking-vs-defensive axis. Forwards and wide attackers (ST, LW, RW, CAM) sit toward positive PC2; center-backs and defensive midfielders (CB, CDM) sit toward negative PC2. Goalkeepers form a separate cluster entirely — expected, since none of the 8 attributes describe goalkeeping. Spread along PC1 within any color reflects overall quality.

4 Insights

- Two components recover interpretable structure — “overall quality” and “attacking vs. defensive orientation” — from 9 raw stats, no labels used.
- PC1 is closer to a proxy for the game’s Overall rating than to any single skill.
- PC2 is the useful axis for archetype-style scouting (attacking vs. defensive profiles).
- Goalkeepers are a structural outlier here — a GK-specific PCA (GK Diving, GK Handling, etc.) would be needed to differentiate them meaningfully.

5 Limitations & Data Notes

- 60 of 18,207 players (0.3%) dropped for missing Position/skill data.
- Only 8 skills + Value used, per the task’s recommended subset; a wider set (Marking, Vision, Positioning) would likely shift loadings, especially PC2.
- Goalkeepers’ outfield-skill values aren’t meaningfully comparable to outfield players’ — visible as a separated cluster, not an error.

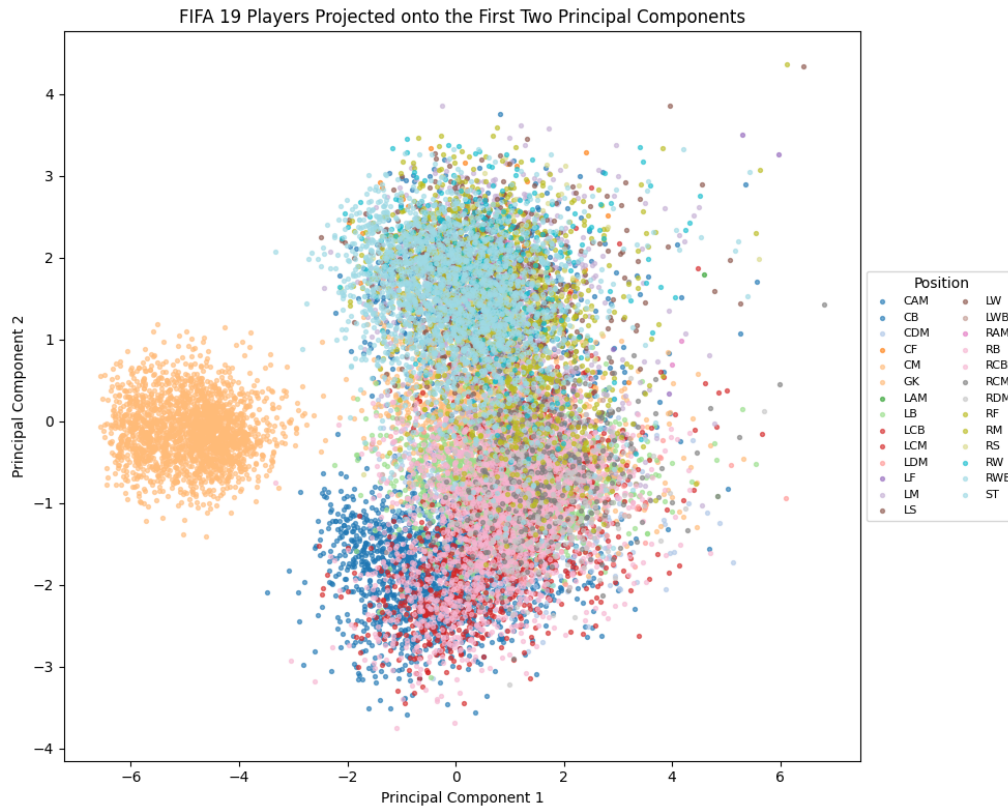


Figure 2: All 18,147 players on PC1/PC2, colored by actual playing position.

- `np.linalg.eig` doesn't fix eigenvector sign — an axis's sign can flip between runs; only relative structure is meaningful.